

Epigenomic Profiling in Neuroblastoma Cell Lines

Kristen Upton¹, Robyn T. Sussman¹, Khushbu Patel², Gregory P. Way³, Gregory I. Sacks¹, Rebecca N. Adams⁴, Paolo Fortina⁴, John M. Maris^{1,5}, Jo Lynne Rokita^{1,2,6}

Affiliations

1. Division of Oncology, Children's Hospital of Philadelphia, Philadelphia, Pennsylvania 19104, USA
2. Department of Bioinformatics and Health Informatics, Children's Hospital of Philadelphia, Philadelphia, Pennsylvania 19104, USA
3. Genomics and Computational Biology Graduate Group, University of Pennsylvania, Philadelphia, PA, 19104, USA
4. Cancer Genomics and Bioinformatics Laboratory, Sidney Kimmel Cancer Center, Philadelphia, Pennsylvania 19107, USA
5. Department of Pediatrics, Perelman School of Medicine at the University of Pennsylvania, Philadelphia, Pennsylvania 19104, USA
6. Center for Data-Driven Discovery in Biomedicine, Children's Hospital of Philadelphia, Philadelphia, Pennsylvania 19104, USA

Corresponding author(s): Jo Lynne Rokita (rokita@email.chop.edu), John Maris (maris@email.chop.edu)

Abstract:

This protocol explains the process of how we collected MYCN, MYC, and Histone ChIP-Seq data, as well as ATAC-Seq data for neuroblastoma cell lines. This protocol is comprised of three sections: Cell Growth and Expansion for care of neuroblastoma cell lines, ChIP-Seq protocol, and ATAC-Seq Protocol. Table 1, within the document, outlines which data was collected for each cell line.

TABLE OF CONTENTS:

- I. [Cell Growth and Expansion](#)
- II. [ChIP-Seq](#)
 - a. [Cell Fixation](#)
 - b. [Chromatin Immunoprecipitation \(ChIP\) and Library Preparation by Active Motif](#)
 - c. [MYCN ChIP-Seq: Kelly and NGP](#)
 - d. [ChIP Library Sequencing by Jefferson Cancer Genomics Laboratory at the Kimmel Cancer Center](#)
 - e. [ChIP-Seq Analysis](#)
- III. [ATAC-Seq](#)
 - a. [Cell Preparation](#)
 - b. [Transposition Reaction and Purification](#)
 - c. [PCR Amplification](#)
 - d. [qPCR](#)
 - e. [Library Preparation](#)
 - f. [Sequencing for ATAC-Seq by Beijing Genomics International \(BGI\)](#)
 - g. [ATAC-Seq NB-69 and NGP via Active Motif](#)
 - h. [ATAC-Seq Analysis](#)
- IV. [References](#)

Table 1. Identification of which assays (ChIP-Seq and/or ATAC-Seq) were performed for each neuroblastoma cell line.

		Histone ChIP-Seq				TF ChIP-Seq		Other
Cell Line	MYCN Status	H3K27Ac	H3K27me3	H3K4me1	H3K4me3	MYCN	MYC	ATAC
Kelly	amp	✓	✓	✓	✓	✓	✓	✓
NGP	amp	✓	✓	✓	✓	✓		✓
COG-N-415	amp	✓	✓	✓	✓	✓		✓
LA-N-5	amp	✓	✓	✓	✓	✓		✓
SK-N-BE2C	amp	✓	✓	✓	✓	✓		✓
NB-1643	amp	✓	✓	✓	✓	✓		✓
NBL-S	non-amp	✓	✓	✓	✓			✓
SK-N-AS	non-amp	✓	✓	✓	✓		✓	✓
SK-N-FI	non-amp	✓	✓	✓	✓			✓
NB-69	non-amp	✓	✓	✓	✓	✓	✓	✓
SK-N-SH	non-amp	*	*	*	*		✓	✓

*External data available from ENCODE

I. Cell Growth and Expansion:

Neuroblastoma cell lines were thawed and cultured using the same materials (see Table 2) and methods as described in Harenza et al., 2017.

Table 2. Cell Line Information: Histology, MYCN status, Growth Media, and Detachment Method

Cell Line	Organism	Cancer Histology	Matched PDX	MYCN Status	Growth Media	Detachment Method
COG-N-415	Homo sapiens	Neuroblastoma	COG-N-415x	Amplified	IMDM, 20% FBS, 2mM L-glutamine, 1:1000 ITS Premix Supplement	0.02% Versene
KELLY	Homo sapiens	Neuroblastoma	N/A	Amplified	RPMI 1640, 10% FBS, 1% Penicillin/Streptomycin, 2mM L-Glutamine	0.02% Versene
LA-N-5	Homo sapiens	Neuroblastoma	N/A	Amplified	RPMI 1640, 10% FBS, 1% Penicillin/Streptomycin, 2mM L-Glutamine	0.02% Versene
NB-1643	Homo sapiens	Neuroblastoma	NB-1643	Amplified	IMDM, 20% FBS, 1% Penicillin/Streptomycin, 2mM L-glutamine	0.02% Versene
NB-69	Homo sapiens	Neuroblastoma	N/A	Single Copy	RPMI 1640, 10% FBS, 1% Penicillin/Streptomycin, 2mM L-Glutamine	0.02% Versene
NBL-S	Homo sapiens	Neuroblastoma	N/A	Single Copy	RPMI 1640, 10% FBS, 1% Penicillin/Streptomycin, 2mM L-Glutamine	0.02% Versene
NGP	Homo sapiens	Neuroblastoma	N/A	Amplified	RPMI 1640, 10% FBS, 1% Penicillin/Streptomycin, 2mM L-Glutamine	0.02% Versene
SK-N-AS	Homo sapiens	Neuroblastoma	N/A	Single Copy	RPMI 1640, 10% FBS, 1% Penicillin/Streptomycin, 2mM L-Glutamine	0.05% Trypsin/EDTA
SK-N-BE2C	Homo sapiens	Neuroblastoma	N/A	Amplified	RPMI 1640, 10% FBS, 1% Penicillin/Streptomycin, 2mM L-Glutamine	0.02% Versene
SK-N-FI	Homo sapiens	Neuroblastoma	N/A	Single Copy	RPMI 1640, 10% FBS, 1% Penicillin/Streptomycin, 2mM L-Glutamine	0.02% Versene

SK-N-SH	Homo sapiens	Neuroblastoma	N/A	Single Copy	RPMI 1640, 10% FBS, 1% Penicillin/Streptomycin, 2mM L-Glutamine	0.05% Trypsin/EDTA
---------	--------------	---------------	-----	-------------	---	--------------------

1. Thaw cells by floating in a 37°C water bath for 2-3 minutes.
2. Add cells to a 15 mL tube containing 5 mL of the appropriate growth media (Table 2), and pellet cells by centrifugation at 300 x *g* for 5 minutes at room temperature.
3. Carefully aspirate off the supernatant to remove the DMSO-containing freezing media and resuspend cells in 1 mL of fresh growth medium.
4. Transfer cells to a T-75 flask containing 10 mL of growth media. Incubate cells at 37°C with 5.0% CO₂ concentration for growth.
5. Once cells are ~70-80% confluent, aspirate off growth media. Gently wash cells with sterile 1X PBS and aspirate off PBS.
6. Add 3 mL of the appropriate detachment solution (Table 2) and incubate at RT or 37°C until cells are detached from the plate (~2-5 min.). Confirm cells are detached by eye (by looking at the bottom of the plate) or using the microscope.
7. Gather cells by tilting the plate at a 45° angle and washing with at least 4 mL appropriate growth media.
8. Collect and transfer cells to a 15 mL conical tube. Centrifuge tube at 300 x *g* for 5 minutes.
9. Aspirate the media and detachment solution, being careful not to disturb the pellet. Resuspend in 1 mL of the appropriate growth medium and transfer cells to a 150 mm dish containing 19 mL of growth media.
10. After reaching ~70-80% confluence, remove cells from the incubator, and pellet or crosslink cells as described.

II. ChIP-Seq Protocol

The ChIP-Seq Protocol is separated into four parts: Cell Fixation, Chromatin Immunoprecipitation (ChIP) and Library Preparation, Library Sequencing, and ChIP-Seq Analysis. Note: MYCN ChIP-Seq for Kelly and NGP was performed using a slightly varied procedure and is noted in a separate section.

Cell Fixation for ChIP-Seq:

Table 3A: Formaldehyde Solution:

	Manufacturer, Catalog#	Final Concentration	Per 20 mL	Per 10 mL
37% Formaldehyde	Sigma, F8775	11%	6 mL	3 mL
5.0 M NaCl	Sigma, S5150	0.1 M	0.4 mL	0.2 mL
0.5 EDTA (pH 8.0)	Invitrogen, 15575-038	1 mM	40 µL	20 µL
1.0 M HEPES (pH 7.9)	Gibco, 15630-080	50 mM	1 mL	0.5 mL
Nuclease-free H ₂ O	Ambion, AM9937		12.6 mL	6.3 mL

Table 3B: Glycine Solution:

Reagents	Manufacturer, Catalog#	Final Concentration	Per 20 mL	Per 10 mL
Glycine (mw 75)	Sigma, G7403	2.5 M	3.75 g	1.875 g
Nuclease-free H ₂ O	Ambion, AM9937		20 mL	10 mL

Cell fixation for chromatin immunoprecipitation was performed on adherent cells as described in the Active Motif,

Epigenetic Services ChIP Cell Fixation Protocol:

1. Freshly prepare the Formaldehyde solution (Table 3A).
2. Add 1/10th of growth media volume to the existing media in each container of cells (e.g. 1/10 volume of 20 mL of media will have 2 mL added). Gently swirl to mix, and rock the plate for 15 minutes at room temperature.
3. Stop the fixation immediately by adding 1/20th volume of the 2.5M Glycine Solution (Table 3B) to the existing media in each plate (e.g. 22 mL of growth media and Formaldehyde Solution would get 1.1 mL of Glycine Solution). Gently swirl to mix, and then allow to sit at room temperature for 5 minutes.
4. After Glycine incubation, hold the plate at a 45° angle and use a cell scraper to collect cells at the bottom edge of the plate. Use a pipette to transfer the cells to a 50 mL conical tube on ice. Keep samples on ice for the remainder of the procedure.

5. Centrifuge the tube(s) at 800 x g at 4°C for 10 minutes to pellet the cells.
6. Remove and discard the supernatant, and resuspend cells in 10 mL of chilled, sterile 1X PBS by pipetting up and down. If there are multiple tubes (e.g. from multiple plates of the same cell line), the samples can be combined at this step.
7. Centrifuge the tube again as before. Remove and discard the supernatant, then add 10 mL of chilled, sterile 1X PBS to the tube. Add 100 µL of 100 mM PMSF to the tube and pipet up and down to resuspend the cells.
8. Centrifuge the tube at 800 x g at 4°C for 10 minutes. Remove and discard the supernatant.
9. Snap-freeze cells in dry ice, and store at -80°C.
10. Ship cells to Active Motif on dry ice following the instructions listed at on the Sample Submission Form, downloaded from www.activemotif.com/sample-submission.

Chromatin Immunoprecipitation (ChIP) and Library Preparation by Active Motif:

Table 4: Antibody information and concentration used for ChIP

Antibody	Manufacturer, Catalog#	Amount of Antibody per Immunoprecipitation (IP)	Chromatin Concentration per IP
N-Myc	Active Motif, #61185	6 µl	30 µg
c-Myc	Santa Cruz N262, #sc-764	20 µl	30 µg
H3K4me1	Active Motif, #39297	5 µl	30 µg
H3K4me3	Active Motif, #39159	3 µl	30 µg
H3K27Ac	Active Motif, #39133	4 µg	30 µg
HeK27me3	Millipore, #07-449	4 µg	30 µg
H3K4me1	Active Motif, #39297	5 µl	30 µg

Methods for chromatin immunoprecipitation and library preparation were conducted by Active Motif. In brief:

1. Chromatin was isolated using a lysis buffer and membranes were disrupted with a dounce homogenizer.
2. Lysates were sonicated with Active Motif's EpiShear probe sonicator (#53051) and cooled sonication platform (#53080) to an average fragment length 300-500 bp.
3. A portion of the sample was collected as the Input, treated with RNase, proteinase K, and incubated to reverse crosslinking. DNA was collected by ethanol precipitation.

4. Input DNA was resuspended and concentration was quantified by a NanoDrop spectrophotometer.
Extrapolation of this concentration to the original chromatin volume allowed for quantitation of the total chromatin yield.
5. Aliquots of fixed chromatin to be used in the immunoprecipitation were precleared with protein A agarose beads (Invitrogen, #15918014).
6. Genomic DNA regions of interest were isolated using specific ChIP antibodies (Table 4).
7. Antibody DNA complexes were isolated using additional beads.
8. The crosslinked DNA, antibody, and bead complexes were washed.
9. The crosslinked DNA was eluted from the beads with an SDS buffer, and subjected to RNase and proteinase K treatment.
10. Reverse crosslinking occurred in an overnight incubation at 65°C, and ChIP DNA was purified with a phenol-chloroform extraction and ethanol precipitation.
11. Illumina sequencing libraries were prepared from the ChIP and Input DNA from the ChIP and Input DNAs using the standard consecutive enzymatic steps of end-polishing, dA-addition, and adaptor ligation using Active Motif's custom liquid handling robotics pipeline.
12. Samplers were amplified with a 15 cycle PCR amplification and then quantified before being shipped to the Jefferson Cancer Genomics Laboratory at the Kimmel Cancer Center for sequencing.

MYCN ChIP-Seq: Kelly and NGP

Chromatin immunoprecipitation was performed on adherent cells as described in Bosse et al., 2017. Of note, a different MYCN antibody was used than listed in Bosse et al., 2017.

Table 5A: FA Lysis Buffer

Reagents	Manufacturer, Catalog#	Final Concentration
1.0 M HEPES (pH 7.5)	Gibco, 15630-080	50 mM
NaCl	Sigma, S5150	140 mM
EDTA (pH 8.0)	Invitrogen, 15575-038	1 mM
Triton-X-100	Sigma, 9002-93-1	0.1%
SDS	Invitrogen, 15553-035	0.1%
Sodium deoxycholate	VWR, 97062-028	0.1%
Protease Inhibitors	Thermo Scientific, 88666	
DTT	Fisher, CAS 348-12-3	1 mM

Table 5B: ChIP Wash Buffer

Reagents	Manufacture, Catalog#	Final Concentration
Tris-HCl (pH 8.0)	Invitrogen, 15568-025	20 mM
NaCl	Sigma, S5150	150 mM
EDTA (pH 8.0)	Invitrogen, 15575-038	2 mM
Triton-X-100	Sigma, 9002-93-1	1.0%
SDS	Invitrogen, 15553-035	0.1%
Sodium deoxycholate	VWR, 97062-028	0.1%

Table 5C: Final Wash Buffer

Reagents	Manufacture, Catalog#	Final Concentration
Tris-HCl (pH 8.0)	Invitrogen, 15568-025	20 mM
NaCl	Sigma, S5150	500 mM
EDTA (pH 8.0)	Invitrogen, 15575-038	2 mM
Triton-X-100	Sigma, 9002-93-1	1.0%
SDS	Invitrogen, 15553-035	0.1%

1. Grow cells as described in Cell Growth and Expansion section of protocol to 70-80% confluence in 150 mm tissue culture plates in 20 mL of media.
2. To the existing media, add 415 μ L of 37% formaldehyde (final concentration of 0.75%), and rock for 10 minutes at room temperature.
3. Add 1.5 mL of 2.5 M glycine (Table 3B) (final concentration of 0.18 M) to inactivate the formaldehyde, and rock plate for an additional 5 min.
4. Lyse cells with a volume of FA Lysis Buffer equivalent to 5 pellet volumes.
5. Wash Protein A beads (Cell Signaling, #9863) 3 times in ChIP Wash Buffer and one time with Final Wash Buffer.
6. Libraries were constructed using NEB Ultra Kit (#E7370S) following manufacturer's instructions.
7. Libraries were sequenced as single-end, 50 bp reads on a MiSeq to a depth of ~50 M reads by the Children's Hospital of Philadelphia Nucleic Acid and PCR Core.

ChIP Library Sequencing for ChIP-Seq by Jefferson Cancer Genomics Laboratory:

Methods for sequencing were conducted by the Jefferson Cancer Genomics Laboratory at the Kimmel Cancer Center. In brief:

1. Samples were quality control tested using an Agilent High Sensitivity Screen Tape to determine average fragment length.
2. The concentration of each library was measured using a High Sensitivity Qubit Quantification kit, and samples were diluted to an appropriate amount for the loading protocol (4 nM or less).
3. Samples were normalized to the same nanomolar concentration, and libraries were pooled together in equal amounts.
4. Samples were diluted to 1.51 pM in Low EDTA TE Buffer.
5. Samples were sequenced as single-end, 70 bp reads to an average depth of ~30 M reads on a NextSeq 500.

ChIP-Seq Analysis

FASTQ quality was assessed using FastQC (<http://www.bioinformatics.babraham.ac.uk/projects/fastqc/>) and sequences were adapter- and quality-trimmed using Trim Galore (F., 2015; Lindgreen, 2012). Since multiple sequencers were used, FASTQ phred sequencing scores (Ewing et al., 1998) were calculated using a perl script (<https://raw.githubusercontent.com/douglasgscfield/bioinfo/master/scripts/phredDetector.pl>). This value was used as input into the alignment algorithm. The bwa samse (Li and Durbin, 2009) was used to align the reads to hg19 reference genome and Picard tools (Alec Wysocki, 2013) was used to remove duplicates. Fragment sizes were estimated using MaSC (Ramachandran et al., 2013) and these values were used as input into MACS2 (Zhang et al., 2008) for narrow peak calling (transcription factors) or broad peak calling (histone marks). Peaks were called significant using a q-value (minimum False Discovery Rate) cut off of 0.05. Results were returned in units of signal per million reads to get normalized peak values. Finally, repetitive centromeric, telomeric and satellite regions known to have low sequencing confidence were removed using the blacklisted regions defined by UCSC.

III. ATAC-Seq Protocol

The following ATAC-Seq protocol is adapted from Buenrostro, et. al, 2015. This protocol consists of four parts:

Cell Preparation, Transposition Reaction and Purification, PCR Amplification, qPCR, and Library Preparation.

Primer 1 and Primer 2 were custom synthesized by Integrated DNA Technologies (IDT), using sequences provided in Buenrostro, et. al, 2015 (Table 7). Note: ATAC-Seq for NB-69 and NGP was performed using a slightly varied procedure and is noted in a separate section.

Table 6A: ATAC-Seq Lysis Buffer

Reagents	Manufacturer, Catalog#	Final Concentration
Tris-HCl (pH 7.4)	Invitrogen, 15567027	10 mM
NaCl	Sigma, S5150	10 mM
MgCl ₂	Fisher, 7791-18-6	3 mM
Igepal CA-630, molecular biology grade	Sigma, I8896	0.1%

Table 6B: Other Reagents

Reagents	Manufacturer, Catalog#	Final Concentration
1X Phosphate Buffered Saline (PBS)	Gibco, 14190-136	10 mM
TDB	Illumina, FC-121-1030 (2X reaction buffer from Nextera Kit)	2X
TDE1	Illumina, FC-121-30 (Nextera Tn5 Transposase from Nextera Kit)	
Nuclease-free H ₂ O	Ambion, AM9937	
NEBNext High-Fidelity 2X PCR Master Mix	NEB, M0541	
10,000X SYBR Green	Invitrogen, S-7563	

Cell Preparation:

1. Grow cells in a T-75 flask as described above.
2. Following detachment and pelleting, resuspend cells in 1.0 mL of the appropriate growth media.
3. Triturate until cells are in a homogenous single-cell suspension.
4. Using an automated cell counter, determine the volume which includes 500,000 cells.

5. Aliquot volume for 500,000 cells into sterile 1.5 mL Eppendorf tube containing 500 μ L of sterile 1X PBS.
6. Centrifuge at 500 x *g* for 5 minutes at 4°C.
7. Carefully aspirate off the supernatant, and resuspend cells in 500 μ L of sterile 1X PBS. Centrifuge at 500 x *g* for 5 minutes at 4°C.
8. Carefully aspirate off the supernatant, and then resuspend cells in 500 μ L of cold lysis buffer by gently pipetting up and down. Immediately centrifuge at 500 x *g* for 10 minutes at 4°C.
9. Carefully remove and discard the supernatant.
10. Resuspend pellet in 50 μ L of nuclease free water by gently pipetting up and down, and continue to Transposition Reaction and Purification immediately.

Transposition Reaction and Purification:

1. Place cell pellet on ice.
2. Make transposition reaction mix, but combining the following:
 - 25 μ L TD (2X reaction buffer from Nextera Kit)
 - 2.5 μ L TDE1 (Nextera Tn5 Transposase from Nextera Kit)
 - 17.5 μ L nuclease-free water
 - 5.0 μ L of resuspended DNA/protein (from Cell Preparation step 15)
3. Incubate transposition reaction in thermocycler at 37°C for 30-35 minutes.
4. Immediately purify using Qiagen MinElute PCR Purification Kit. Elute transposed DNA in 10.5 μ L of elution buffer (Buffer EB from the MinElute Kit consisting of 10 mM Tris-Cl (pH 8)).
5. Parafilm Eppendorf tube containing purified DNA and store at -20°C if necessary. (*Note: This can act as a good stopping point, however these DNA fragments are not PCR amplifiable if melted at this point.*)

PCR Amplification:

Table 7: Primer Sequences used for amplification

Primer Name	Sequence	Concentration	Purification
-------------	----------	---------------	--------------

Primer 1 (Ad1)	AATGATACGGCGACCACCGAGATCTACACTCGTCGGCAGCGTCAGATGTG	25 nm	STD
Primer2.1_TAAGGCGA	CAAGCAGAAGACGGCATACGAGATTCGCCTTAGTCTCGTGGGCTCGGAGATGT	25 nm	STD
Primer2.2_CGTACTAG	CAAGCAGAAGACGGCATACGAGATCTAGTACGGTCTCGTGGGCTCGGAGATGT	25 nm	STD
Primer2.3_AGGCAGAA	CAAGCAGAAGACGGCATACGAGATTCTGCCTGTCTCGTGGGCTCGGAGATGT	25 nm	STD
Primer2.4_TCCTGAGC	CAAGCAGAAGACGGCATACGAGATGCTCAGGAGTCTCGTGGGCTCGGAGATGT	25 nm	STD
Primer2.5_GGACTCCT	CAAGCAGAAGACGGCATACGAGATAGGAGTCCGTCTCGTGGGCTCGGAGATGT	25 nm	STD
Primer2.6_TAGGCATG	CAAGCAGAAGACGGCATACGAGATCATGCCTAGTCTCGTGGGCTCGGAGATGT	25 nm	STD
Primer2.7_CTCTCTAC	CAAGCAGAAGACGGCATACGAGATGTAGAGAGGTCTCGTGGGCTCGGAGATGT	25 nm	STD
Primer2.8_CAGAGAGG	CAAGCAGAAGACGGCATACGAGATCCTCTCTGGTCTCGTGGGCTCGGAGATGT	25 nm	STD
Primer2.9_GCTACGCT	CAAGCAGAAGACGGCATACGAGATAGCGTAGCGTCTCGTGGGCTCGGAGATGT	25 nm	STD
Primer2.10_CGAGGCTG	CAAGCAGAAGACGGCATACGAGATCAGCCTCGGTCTCGTGGGCTCGGAGATGT	25 nm	STD
Primer2.11_AAGAGGCA	CAAGCAGAAGACGGCATACGAGATTGCCTCTTGTCTCGTGGGCTCGGAGATGT	25 nm	STD
Primer2.12_GTAGAGGA	CAAGCAGAAGACGGCATACGAGATTCCTCTACGTCTCGTGGGCTCGGAGATGT	25 nm	STD
Primer2.13_GTCGTGAT	CAAGCAGAAGACGGCATACGAGATATCACGACGTCTCGTGGGCTCGGAGATGT	25 nm	STD
Primer2.14_ACCACTGT	CAAGCAGAAGACGGCATACGAGATACAGTGGTGTCTCGTGGGCTCGGAGATGT	25 nm	STD
Primer2.15_TGGATCTG	CAAGCAGAAGACGGCATACGAGATCAGATCCAGTCTCGTGGGCTCGGAGATGT	25 nm	STD
Primer2.16_CCGTTTGT	CAAGCAGAAGACGGCATACGAGATACAAACGGGTCTCGTGGGCTCGGAGATGT	25 nm	STD
Primer2.17_TGCTGGGT	CAAGCAGAAGACGGCATACGAGATACCCAGCAGTCTCGTGGGCTCGGAGATGT	25 nm	STD
Primer2.18_GAGGGGTT	CAAGCAGAAGACGGCATACGAGATAACCCCTCGTCTCGTGGGCTCGGAGATGT	25 nm	STD
Primer2.19_AGTTGGG	CAAGCAGAAGACGGCATACGAGATCCCAACCTGTCTCGTGGGCTCGGAGATGT	25 nm	STD
Primer2.20_GTGTGGTG	CAAGCAGAAGACGGCATACGAGATCACCACAGTCTCGTGGGCTCGGAGATGT	25 nm	STD
Primer2.21_TGGGTTTC	CAAGCAGAAGACGGCATACGAGATGAAACCCAGTCTCGTGGGCTCGGAGATGT	25 nm	STD
Primer2.22_TGGTCACA	CAAGCAGAAGACGGCATACGAGATTGTGACCAGTCTCGTGGGCTCGGAGATGT	25 nm	STD
Primer2.23_TTGACCCT	CAAGCAGAAGACGGCATACGAGATAGGGTCAAGTCTCGTGGGCTCGGAGATGT	25 nm	STD
Primer2.24_CCACTCCT	CAAGCAGAAGACGGCATACGAGATAGGAGTGGGTCTCGTGGGCTCGGAGATGT	25 nm	STD

1. To amplify Transposed DNA, combine the following into 0.2 mL PCR tube:

- 10 µL transposed DNA
- 10 µL nuclease-free H₂O
- 2.5 µL 25 µM PCR Primer 1 (Ad1)
- 2.5 µL 25 µM Barcoded PCR Primer 2 (Ad2.X, X meaning the number of unique samples)
- 25 µL NEBNext High-Fidelity 2X PCR Master Mix

2. Thermal cycle as follows:

1 cycle:	5 min 72°C
	30 sec 98°C
5 cycles:	10 sec 98°C
	30 sec 63°C
	1 min 72°C

The five minute extension is critical to allow extension on both ends of the primer after transposition, thereby generating amplifiable fragments. This ensures that downstream quantitative PCR (qPCR) quantitation will not change the complexity of the original library.

qPCR:

Note: To reduce the GC and size bias in PCR, the appropriate number of PCR cycles (N) is determined using qPCR, allowing us to stop prior to saturation.

1. Keep the samples in the thermocycler following the PCR Amplification reaction and run a qPCR side reaction:

- 5 µL of DNA PCR amplified DNA
- 2 µL of nuclease free H₂O
- 1 µL of 6.25 µM Custom Nextera PCR Primer (Ad1)
- 1 µL of 6.25 µM Custom Nextera PCR Primer 2 (Ad2.X)
- 1 µL 9X SYBR Green I
- 5 µL NEBNext High-Fidelity 2X PCR Master Mix

2. Cycles on the qPCR instrument are as follows:

1 cycle:	30 sec 98°C
20 cycles:	10 sec 98°C
	30 sec 63°C
	1 min 72°C

3. To calculate the additional number of cycles needed, plot linear R_n versus cycle and determine the cycle number that corresponds to one-third of the maximum fluorescent intensity.

4. Run the remaining 45 μ L PCR reaction to the cycle number determined by qPCR. Cycles are as follows:

1 cycle:	30 sec 98°C
N cycles:	10 sec 98°C
	30 sec 63°C
	1 min 72°C
6. Purify the amplified library using Qiagen MinElute PCR Purification Kit. Elute purified library in 20 μ L of elution buffer (Buffer EB from the MinElute Kit consisting of 10 mM Tris-Cl (pH 8)). Make sure that the column is dry prior to adding elution buffer to avoid ethanol contamination of final library. (*Note: Sample can be stored at -20°C at this point.*)
7. Purify amplified library using AMPure XP beads at a 1.8x ratio to get rid of adapter dimer, using 80% ethanol for the wash steps. Elute in 50 μ L of nuclease free H₂O. (*Note: Sample can be parafiled and stored at -20°C at this point, if needed.*)
8. Concentration of the DNA eluted from the column should be about 30 nM.

Library Preparation

1. Assess the quality of purified libraries using a Bioanalyzer High-Sensitivity DNA Analysis kit (Agilent).
2. If large, predominant peak exists around 1000 bp, remove with SPRI beads.
 - First, with a new vial of SPRI beads, perform size selection with various ratios to ensure large peak is removed. For example, 0.4X, 0.45X, 0.5X. Choose the ratio that removes 1000 bp fragments, but leaves 800 bp fragments.
 - Elute in 20 μ L of nuclease-free water.

Note: Samples can be parafiled and stored at -20°C at this point.

3. Libraries were sequenced by Beijing Genomics International at the Children's Hospital of Philadelphia.

Sequencing for ATAC-Seq by Beijing Genomics International (BGI):

Methods for sequencing were conducted by Beijing Genomics International at the Children's Hospital of Philadelphia. In brief:

1. Samples were quality control tested using an Agilent High Sensitivity Screen Tape to confirm average fragment sizes were ~180, 380, 580, 780, 980 bp.
2. The concentration of each library was measured using a High Sensitivity Qubit Quantification kit, to ensure they were ≥ 5.5 nM.
3. Samples were normalized and libraries were pooled together in equal amounts.
4. Samples were sequenced as paired-end, 100 bp to an average depth of 80M reads on a HiSeq 2500.

ATAC-Seq for NB-69 and NGP via Active Motif:

1. Grow cells in a T-75 cm² flask as described above.
2. Following detachment and pelleting, resuspend cells in 1.0 mL of the appropriate growth media.
3. Triturate until cells are in a homogenous single-cell suspension.
4. Using an automated cell counter, determine the volume which includes 100,000 cells.
5. Aliquot volume for 100,000 cells into a sterile 1.5 mL Eppendorf tube containing 500 μ L of sterile 1X PBS.
6. Centrifuge at 500 x g for 5 minutes at 4°C.
7. Carefully aspirate off the supernatant, and resuspend cells in 500 μ L of growth media plus 5% DMSO and transfer to a 1.7 mL microfuge tube on ice.
8. Freeze the cells at a slow cooling to minimize cell lysis. A Mr. Frosty (Thermo Scientific, 5100-0001) container was used in this preparation.
11. Ship samples on dry ice to Active Motif (1914 Palomar Oaks Way, Ste 150, Carlsbad, CA 92008) following the instructions listed at on the Sample Submission Form, downloaded from www.activemotif.com/sample-submission.
12. Samples were prepared and sequenced following Active Motif's ATAC-Seq proprietary protocol. If brief:
 - a. Cells were thawed in a 37°C water bath, pelleted, washed with cold PBS, and tagmented as previously described (Buenrostro et al. 2013), with some modifications based on (Corces et al. 2017).

- b. Cell pellets were resuspended in lysis buffer, pelleted, and tagmented using the enzyme buffer provided in the Nextera Library Prep Kit (Illumina).
- c. Tagmented DNA was then purified using the MinElute PCR purification kit (Qiagen), amplified with 10 cycles of PCR, and purified using Agencourt AMPure SPRI beads (Beckman Coulter).
- d. The resulting material was quantified using the KAPA Library Quantification Kit for Illumina platforms (KAPA Biosystems) and sequenced with PE42 sequencing on the NextSeq 500 sequencer (Illumina).

ATAC-Seq Analysis

Samples were quality-controlled and trimmed as described in Chip-Seq Analysis. FASTQ files were aligned using `bwa aln` for BGI samples (100 bp reads) and `bwa mem` for Active Motif samples (42 bp reads). Reads with mapping quality < 10 were discarded. Biological duplicate BAMs were merged using Picard tools. Broad peaks were called using `--extsize 200` and `--shift 100`. Peaks were called significant using a q-value (minimum False Discovery Rate) cut off of 0.05. Results were returned in units of signal per million reads to get normalized peak values. Finally, repetitive centromeric, telomeric and satellite regions known to have low sequencing confidence were removed using the blacklisted regions defined by UCSC.

References

- Bosse, K.R., Raman, P., Zhu, Z., Lane, M., Martinez, D., Heitzeneder, S., Rath, K.S., Kendersky, N.M., Randall, M., Donovan, L., *et al.* (2017). Identification of GPC2 as an Oncoprotein and Candidate Immunotherapeutic Target in High-Risk Neuroblastoma. *Cancer Cell*, 32, 295-309 e212.
- Buenrostro, J., Wu, B., Chang, H., and Greenleaf, W. ATAC-Seq: A Method for Assaying Chromatin Accessibility Genome Wide. *Curr Protoc Mol Biol*, 2015; 109: 21.29.1–21.29.9.
- Corces et al. An improved ATAC-seq protocol reduces background and enables interrogation of frozen tissues. *Nat Methods*, 2017, 14(10): 959-962.
- Harenza, J.L., Diamond, M.A., Adams, R.N., Song, M.M., Davidson, H.L., Hart, L.S., Dent, M.H., Fortina, P., Reynolds, C.P., and Maris, J.M. (2017). Transcriptomic profiling of 39 commonly-used neuroblastoma cell lines. *Sci Data*, 4, 170033.